

LIPA CART

User Experience Specification Document

Version 1.0

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1. Executive Summary

LIPACART is a mobile-first grocery shopping and delivery platform that connects customers with personal shoppers at local markets and boda boda delivery riders. The app streamlines the grocery shopping experience by enabling users to create shopping lists, make secure payments, and receive fresh groceries delivered directly to their doorstep.

1.1 Design Philosophy

- **Simplicity First:** The interface prioritises ease of use for users with varying levels of digital literacy. Every interaction should be intuitive and require minimal steps to complete.
- **Trust & Transparency:** Real-time updates, clear pricing, and transparent processes build confidence in the service. Users can track their orders from shopping to delivery.
- **Local-First Approach:** The design celebrates local markets and community commerce while bringing modern convenience to traditional shopping habits.
- **Accessibility:** The app accommodates users across various demographics, including those in areas with limited internet connectivity, through optimised data usage and offline capabilities.
- **Brand Identity:** The orange and green colour scheme reflects freshness (green) and energy/enthusiasm (orange), creating a vibrant, approachable brand that feels both modern and grounded in local culture.

2. Target Users

2.1 Primary Users (Customers)

Profile: Urban professionals, busy parents, elderly individuals, and anyone seeking convenient grocery shopping

Characteristics:

- Age range: 25-55 years old (primary), with secondary users 18-70+
- Tech comfort level: Basic to intermediate smartphone users
- Income level: Middle income and above
- Pain points: Time constraints, difficulty visiting markets, transportation challenges, physical mobility limitations

- Goals: Save time, get fresh produce, avoid market crowds, and have convenient payment options

User Needs:

- Quick list creation with common items readily available
- Transparent pricing before payment
- Reliable delivery timing
- Quality assurance on fresh produce
- Multiple payment options (mobile money, credit cards)

2.2 Secondary Users (Personal Shoppers)

Profile: Market vendors or designated shoppers with local market knowledge

Characteristics:

- Age range: 20-50 years old
- Location: Based at or near local markets
- Tech comfort level: Basic smartphone proficiency required
- Goals: Earn income, efficient order management, and clear communication with customers
- Pain points: Managing multiple orders, payment delays, and unclear item specifications

User Needs:

- Clear order details with quantities and customer preferences
- Easy communication channel with customers for substitutions
- Simple order completion workflow
- Reliable payment receipt confirmation

2.3 Tertiary Users (Boda Boda Riders)

Profile: Motorcycle taxi operators providing delivery services

Characteristics:

- Age range: 20-45 years old
- Tech comfort level: Basic smartphone and GPS usage
- Goals: Maximise deliveries, minimise wait time, clear delivery instructions
- Pain points: Incorrect addresses, customer unavailability, payment disputes

User Needs:

- Clear pickup and delivery locations with GPS integration
- Customer contact information
- Delivery proof mechanism
- Fair compensation tracking

3. Core Features

3.1 Customer App Features

Onboarding & Authentication

- Quick registration with phone number verification
- Optional social media login
- Profile setup with delivery addresses and payment methods
- Brief tutorial highlighting key features

Shopping List Creation

- Smart search with autocomplete for common items
- Category browsing (Vegetables, Fruits, Meat, Dairy, Pantry, etc.)
- Quick add buttons for frequently purchased items
- Quantity specification with common units (kg, pieces, bunches)
- Special instructions field per item
- Save lists as templates for recurring purchases
- Share lists with family members

Pricing & Payment

- Estimated total before checkout
- Service fee and delivery fee transparency
- Multiple payment options: Mobile Money (MTN, Airtel), Visa, Mastercard
- Saved payment methods for quick checkout
- Payment receipt via SMS and in-app

Order Tracking

- Real-time status updates: Order received → Shopping in progress → Ready for delivery → In transit → Delivered
- Shopper assigned notification with profile

- Live updates from the shopper with photos of items if needed
- Estimated delivery time
- Boda boda rider tracking on a map
- Push notifications at each status change

Communication

- In-app chat with the assigned shopper for item clarifications
- Call option to the shopper or rider
- Substitution approval system (if item unavailable)

Order History & Management

- Past orders with the reorder option
- Ratings and reviews for shoppers and riders
- Digital receipts with itemised costs
- Dispute resolution system

3.2 Shopper App Features

Order Management Dashboard

- Incoming order notifications
- Accept/decline order capability
- Order details view with shopping list
- Customer's special instructions are prominently displayed
- Timer showing expected shopping completion time

Shopping Workflow

- Item-by-item checklist
- Mark items as found/unavailable
- Price entry per item (if not pre-priced)
- Photo upload capability for customer approval
- Substitution suggestion system
- Item notes from the customer highlighted

Customer Communication

- Quick message templates for common situations
- Chat function for specific questions
- Voice call option for urgent clarifications

Order Completion

- Total cost calculation review
- Package photo for documentation
- Mark as ready for pickup
- Handoff confirmation to the boda boda rider

3.3 Boda Boda Rider App Features

Delivery Management

- Available delivery notifications
- Accept delivery with pickup location and delivery address
- GPS navigation to the pickup point (market/shopper)
- GPS navigation to customer's delivery address

Delivery Workflow

- Order pickup confirmation with photo
- Customer contact information is readily accessible
- Arrival notification to the customer
- Delivery proof: Customer signature or photo
- Payment confirmation (if the cash on delivery option exists)

Earnings Tracking

- Completed deliveries log
- Earnings per delivery
- Weekly/monthly earnings summary

3.4 Admin Dashboard (Web-based)

Operations Management

- Real-time order monitoring across all stages
- Shopper and rider performance metrics
- Customer service inquiry management
- Dispute resolution interface

Analytics & Reporting

- Order volume trends

- Popular items and categories
- Peak ordering times
- Customer retention rates
- Average delivery times
- Revenue analytics

User Management

- Customer account administration
- Shopper onboarding and verification
- Rider onboarding and verification
- Rating and review moderation

4. Technical Specifications

4.1 User Interface Design System

Color Palette

- Primary Orange: **#FF8C00** (energetic, action buttons, highlights)
- Primary Green: **#2ECC71** (fresh, success states, confirmation)
- Secondary Orange: **#FFA500** (lighter variant for backgrounds)
- Secondary Green: **#27AE60** (darker variant for text on light backgrounds)
- Neutral Grey: **#95A5A6** (borders, inactive states)
- Dark Text: **#2C3E50** (primary text)
- Light Background: **#FFFFFF**
- Error Red: **#E74C3C** (alerts, errors)

Typography

- Primary Font: Inter or Roboto (highly legible, multi-language support)
- Headings: Bold, 24-32px
- Body Text: Regular, 16-18px (larger for accessibility)
- Buttons: Semi-bold, 16-18px
- Captions: Regular, 14px

Component Library

- Buttons: Rounded corners (8px), high contrast, minimum touch target 44x44px
- Input Fields: Clear labels, validation states, helper text

- Cards: Subtle shadows, 12px border radius, organised content grouping
- Lists: Clear separators, swipe actions where appropriate
- Navigation: Bottom tab bar for main sections, top app bar for context
- Modals: Centre-screen for important decisions, bottom sheets for options
- Icons: Line style or filled, consistent weight, intuitive meanings

Spacing & Layout

- Base unit: 8px grid system
- Content padding: 16px minimum from screen edges
- Element spacing: 8px, 16px, 24px for varying hierarchy
- Maximum content width on tablets: 600px (centred)

4.2 Platform Requirements

Backend & Database

- Database: Supabase (PostgreSQL)
- Real-time: Supabase Real-time subscriptions
- Storage: Supabase Storage
- Authentication: Supabase Auth

Mobile Apps (Customer, Shopper, Rider)

- Framework: Flutter
- Language: TypeScript
- Why:
 - Write once, deploy to iOS & Android
 - Large community & packages

Admin Dashboard

- Framework: Next.js 14 (React)
- Language: TypeScript
- UI: Tailwind CSS + shadcn/ui
- Why:
 - Fast development
 - Great Supabase integration
 - SEO-friendly if needed
 - Easy deployment (Vercel)

Payment Integration

- **Mobile Money:** MTN MoMo API, Airtel Money API
- **Cards:** Flutterwave or Paystack

Maps & Location

- Google Maps API or Mapbox

4.3 Performance Requirements

Speed

- App launch time: < 3 seconds
- Page load time: < 2 seconds on 3G connection
- Search results: < 1 second
- Image loading: Progressive with placeholders

Offline Capability

- Shopping list creation and editing offline (syncs when online)
- View past orders offline
- Cached product catalogue for browsing

Data Usage

- Optimised images (WebP format, compressed)
- Minimal data transfer per session (< 5MB for typical browsing)
- Option to disable image auto-loading on slow connections

4.4 Localization

Language Support

- English (primary)
- Local languages as needed (Swahili, Luganda, etc.)
- Right-to-left language support architecture

Currency & Units

- Local currency display
- Common local measurement units (kg, litres, pieces)

- Conversion calculator for units

5. Key Benefits

5.1 For Customers

- **Time Savings:** Eliminate travel time to markets and reduce shopping time from hours to minutes. Users can place orders during commute, lunch breaks, or while managing other tasks.
- **Convenience:** Shop 24/7 from anywhere with delivery to your doorstep. Particularly valuable during bad weather, illness, or when caring for children.
- **Fresh Quality:** Direct sourcing from local markets ensures fresh produce comparable to shopping in person. Photo verification allows quality checks before acceptance.
- **Transparent Pricing:** No hidden costs. See exact prices, service fees, and delivery costs upfront before payment confirmation.
- **Flexible Payment:** Multiple payment options, including mobile money (widespread in East Africa) and international credit cards cater to diverse user preferences.
- **Accessibility:** Elderly users, people with disabilities, or those without transportation can access fresh market produce easily.

5.2 For Personal Shoppers

- **Income Opportunity:** Generate earnings by leveraging existing market knowledge and location. Flexible working hours around existing vendor activities.
- **Simple Technology:** App designed for easy use, even with basic smartphone skills. Minimal training required to get started.
- **Order Management:** an organised system replaces chaotic phone calls and text messages. Clear instructions reduce errors and customer disputes.
- **Payment Security:** Guaranteed payment upon order completion eliminates the risk of non-payment common in informal arrangements.
- **Customer Base Growth:** Access to customers beyond physical market reach, expanding business opportunities.

5.3 For Boda Boda Riders

- **Consistent Work:** Regular delivery requests supplement the traditional passenger transport business, especially during off-peak hours.

- **Fair Pricing:** Transparent delivery fee structure ensures fair compensation based on distance and time.
- **Route Optimisation:** GPS navigation reduces time spent finding locations, allowing more deliveries per day.
- **Reliable Earnings:** Weekly or daily payment options provide financial predictability.
- **Safety:** Reduced haggling over fares and formal system documentation provides security.

5.4 For the Business

- **Scalable Model:** Digital platform enables rapid expansion to new markets and cities with minimal physical infrastructure.
- **Data-Driven Insights:** Customer behaviour, popular products, and demand patterns inform business decisions and partnerships.
- **Marketplace Potential:** Platform can expand to partner with specific vendors, offer bulk discounts, or introduce premium services.
- **Community Impact:** Supports local markets and creates employment for shoppers and riders, contributing to the local economy.
- **Low Overhead:** Asset-light model without inventory or physical stores keeps costs manageable.

6. Security, Integration & Deployment

6.1 Security Measures

Data Protection

- End-to-end encryption for all data transmission (TLS 1.3+)
- Personal information is encrypted at rest in the database
- PCI DSS compliance for credit card processing
- Regular security audits and penetration testing
- GDPR/local data protection regulation compliance

Authentication & Authorization

- Phone number verification via OTP (One-Time Password)
- Biometric authentication option (fingerprint, Face ID) for app access
- Secure password requirements (minimum 8 characters, mixed case, numbers)

- Two-factor authentication for admin accounts
- Role-based access control for shopper, rider, and admin functions
- Automatic session timeout after inactivity

Payment Security

- Tokenisation of payment methods (no raw card numbers stored)
- Integration with PCI-compliant payment gateways
- Transaction monitoring for fraud detection
- Secure API keys management (not hardcoded in apps)
- Payment confirmation via SMS and push notification

Privacy

- Clear privacy policy presented during onboarding
- Granular permissions for location, camera, and contacts
- Option to delete account and associated data
- Anonymous analytics where possible
- Customer address visible only to the assigned shopper/rider for the active order

6.2 Third-Party Integrations

Payment Gateways

- Mobile Money: MTN Mobile Money API, Airtel Money API
- Card Processing: Stripe, Flutterwave, or Paystack for African markets
- Webhook integration for real-time payment status updates

Mapping & Location Services

- Google Maps API or Mapbox for GPS, routing, and location search
- Geocoding for address verification
- Distance calculation for delivery fee computation
- Real-time traffic data for delivery time estimates

Communication Services

- Twilio or Africa's Talking for SMS notifications
- Firebase Cloud Messaging or OneSignal for push notifications
- In-app calling via Twilio Voice or similar VoIP services
- WhatsApp Business API integration (future consideration)

Analytics & Monitoring

- Google Analytics or Mixpanel for user behaviour tracking
- Crashlytics or Sentry for error monitoring and app stability
- Performance monitoring tools (New Relic, Datadog)

Cloud Services

- AWS S3 or Cloudinary for image storage and optimisation
- AWS CloudFront or similar CDN for fast content delivery
- Cloud Functions for serverless backend processing

6.3 Deployment Strategy

Development Phases

Phase 1: MVP Launch (Months 1-3)

- Core customer app: List creation, payment, basic tracking
- Shopper app: Order receiving, shopping workflow, completion
- Rider app: Delivery management basics
- Single city pilot (e.g., Kampala, Nairobi, or Lagos)
- Manual admin oversight via web dashboard
- Focus on user feedback and rapid iteration

Phase 2: Enhancement (Months 4-6)

- Advanced features: Saved lists, item photos, substitution system
- Improved tracking and notifications
- Rating and review system
- Customer support chat integration
- Performance optimisation based on usage data
- Expand to 2-3 additional neighbourhoods/zones

Phase 3: Scaling (Months 7-12)

- Multi-city expansion
- Rider and shopper recruitment tools
- Advanced analytics dashboard
- Promotional system (discounts, referral program)
- Partnership with specific vendors or markets
- API for potential B2B integrations

Testing Strategy

- Unit testing for critical business logic (payment, order processing)
- Integration testing for API endpoints
- User acceptance testing with the target demographic
- Beta testing program with early adopters in each user category
- Load testing to ensure the system handles peak demand
- Security testing by a third-party firm before public launch

Release Management

- Staged rollout: Internal team → Beta testers → Limited public → Full release
- Feature flags for gradual feature introduction
- A/B testing for UI/UX variations
- Hotfix process for critical bugs
- Regular update cycles (bi-weekly or monthly)

Infrastructure

- Auto-scaling backend to handle demand spikes
- Database replication for reliability
- Automated backups (daily full, hourly incremental)
- Disaster recovery plan with RTO < 4 hours
- Multi-region deployment for future international expansion

Monitoring & Maintenance

- 24/7 system monitoring with alert escalation
- Performance dashboards for real-time health checks
- Weekly code reviews and technical debt assessment
- Monthly security patches and dependency updates
- Quarterly infrastructure review and optimisation

Conclusion

LIPACART represents a comprehensive solution that modernises grocery shopping while supporting local market ecosystems. This specification provides the foundation for building a user-centred platform that prioritises simplicity, trust, and accessibility across all user types. Success will be measured through user adoption rates, order completion rates, customer satisfaction scores, and positive economic impact on shoppers and riders.

The phased approach allows for validated learning and iteration based on real user behaviour, ensuring the product truly serves its community while maintaining financial sustainability.